

Exploratory Data Analysis with Alzheimer's and Dementia

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STAT 490: Special Topics in Statistics

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May 19, 2025

Introduction

Across the globe, about 55 million individuals are living with Alzheimer's or other types of dementias. Dementia is a mental degradation of memory, language, problem-solving and other thinking abilities. Symptoms of those who suffer from dementia are severe to the point where they interfere with everyday life. There are many types of dementia, and Alzheimer's is one of the most common. Alzheimer's is a disease that negatively affects an individual's memory and behavior.

Due to the fact that Alzheimer's and dementia are interrelated, this raises the question of is there a way to use one to predict the chance of having the other? Can we use previous medical data from dementia and Alzheimer's patients to gain knowledge about whether other individuals will have those same diseases? This exploratory data analysis study seeks to use previous medical data from patients with Alzheimer's dementia, or both to predict if other individuals, based on their own medical data, will suffer from Alzheimer's, dementia, or both. The goal of this analysis is to determine if we can accurately predict whether an individual will have Alzheimer's, dementia, or both based on known factors about their health. Additionally, this study seeks to discover any statistically significant relationships between the changes of having both Alzheimer's and dementia.

Exploratory Data Analysis

The dataset used for this analysis will be from Kaggle. This data is provided by The Open Access Series of Imaging Studies (OASIS). The dataset includes informational column of subject ID, MRI ID, group (demented or nondemented), visit order, MR delay time, gender, right-handed

or left-handed, age, years of education, socioeconomic status, mini mental state examination, clinical dementia rating, estimated, total intracranial volume, and normalize whole brain volume.

Models and Methods

The primary responses that this exploratory data analysis study will use are Mini-Mental State Examination (MMSE) score for Alzheimer diagnosis and Clinical Dementia Rating (CDR) for dementia. Both of these variables are ordinal and observed over time.

For the MMSE variable, the range is 0-30, with four diagnosis groups of normal cognition (24-30), mild dementia (19-23), moderate dementia (10-18), and severe dementia (0-9). These categories will be used to create an “Alz” variable. This variable will range from 0-3, with zero as normal cognition, one as mild Alzheimer’s, two as moderate Alzheimer’s, and three as severe forms of Alzheimer’s. Additionally, a binary response variable denoted at “Alz01” will be created. This variable will contain zero if there is no disease and one if the subject has mild to severe forms of Alzheimer’s.

For the CDR variable, this variable ranges from 0-3, with zero as no dementia, one as mild dementia, two as moderate dementia, and three as severe forms of dementia. Additionally, a binary response variable denoted at “CDR01” will be created. This variable will contain zero if there is no disease and one if the subject has mild to severe forms of dementia.

Once these variables are organized and created, models will be built to begin the data analysis. This data analysis study will be using general linear models to explore this dataset. In all the models, the effect of socio-economic status on the disease status will be tested at the baseline and on disease progression.

Training sets of the data and testing sets of the data will be created and run, and the test data will contain a random twenty percent. Firstly, predictive models will be made for predicting the values of “Alz01” and “CDR01” for the test sets with models built on the training sets. These models will be of the general linear model family. Secondly, predictive models will be made for predicting the values of “Alz” and “CDR” for the test sets with models built on the training sets. These models will also be of the general linear model family.

Using the test data, we will predict new values for “Alz01” and “CDR01”, and then for “Alz” and “CDR”. Once this is complete, we will explore the correlation between “Alz01” and “CDR01”, and then for “Alz” and “CDR”. Confusion matrices with the accuracy percentages for these four variables will be found as well.

Results

Here we will observe the results from the general linear models:

Output from the predictive model from the trained data for “Alz01”:

```
Call:
glm(formula = Alz01 ~ Visit + MR.Delay + Age + EDUC + eTIV +
     SES + CDR + nWBV + ASF, family = binomial, data = alz01_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.77578  -0.20347  -0.08256  -0.03646   2.98082

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  63.735737  34.569844   1.844 0.065230 .
Visit         0.976184   1.119460   0.872 0.383201
MR.Delay     -0.002149   0.001883  -1.141 0.253722
Age          -0.145487   0.047813  -3.043 0.002344 **
EDUC         -0.105137   0.150915  -0.697 0.486014
eTIV         -0.008927   0.011079  -0.806 0.420366
SES           0.626252   0.393475   1.592 0.111476
CDR           4.713198   0.888319   5.306 1.12e-07 ***
nWBV        -43.357559  12.055425  -3.597 0.000323 ***
ASF          -11.274128  14.105259  -0.799 0.424125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 227.243  on 283  degrees of freedom
Residual deviance:  92.809  on 274  degrees of freedom
AIC: 112.81

Number of Fisher Scoring iterations: 7
```

At the 0.05 significance level, we can observe that the variables of Age, CDR, and nWBV are significant in the model.

Output from the predictive model from the trained data for “CDR01”:

```
Call:
glm(formula = CDR01 ~ Visit + MR.Delay + Age + EDUC + eTIV +
     SES + Alz01 + nWBV + ASF, family = binomial, data = cdr01_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.77173  -0.27301  -0.15357  -0.08414   2.77850

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  23.981771  36.598993   0.655   0.5123
Visit         2.062159   1.106583   1.864   0.0624 .
MR.Delay     -0.003799   0.001826  -2.080   0.0375 *
Age          -0.057687   0.042836  -1.347   0.1781
EDUC         0.040930   0.146684   0.279   0.7802
eTIV        -0.001012   0.011624  -0.087   0.9306
SES          -0.381193   0.392745  -0.971   0.3318
Alz01         3.655813   0.672200   5.439 5.37e-08 ***
nWBV        -28.217863  11.412704  -2.472   0.0134 *
ASF          -2.383264  15.424158  -0.155   0.8772
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 195.815  on 283  degrees of freedom
Residual deviance:  97.812  on 274  degrees of freedom
AIC: 117.81

Number of Fisher Scoring iterations: 7
```

At the 0.05 significance level, we can observe that the variables of Visit, MR.Delay, Alz01, and nWBV are significant in the model.

Output from the predictive model from the trained data for “Alz”:

```
Call:
glm(formula = Alz ~ Visit + MR.Delay + Age + EDUC + eTIV + SES +
     CDR + nWBV + ASF, family = binomial, data = alz_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.77578  -0.20347  -0.08256  -0.03646   2.98082

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  63.735737  34.569844   1.844 0.065230 .
Visit         0.976184   1.119460   0.872 0.383201
MR.Delay     -0.002149   0.001883  -1.141 0.253722
Age          -0.145487   0.047813  -3.043 0.002344 **
EDUC         -0.105137   0.150915  -0.697 0.486014
eTIV         -0.008927   0.011079  -0.806 0.420366
SES           0.626252   0.393475   1.592 0.111476
CDR           4.713198   0.888319   5.306 1.12e-07 ***
nWBV        -43.357559  12.055425  -3.597 0.000323 ***
ASF          -11.274128  14.105259  -0.799 0.424125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 227.243  on 283  degrees of freedom
Residual deviance:  92.809  on 274  degrees of freedom
AIC: 112.81

Number of Fisher Scoring iterations: 7
```

At the 0.05 significance level, we can observe that the variables of Age, CDR, and nWBV are significant in the model.

Output from the predictive model from the trained data for “CDR”:

```
Call:
glm(formula = CDR ~ Visit + MR.Delay + Age + EDUC + eTIV + SES +
     Alz01 + nWBV + ASF, data = cdr01_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.4280  -0.1947  -0.1063   0.2306   1.9120

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.411e+00  2.208e+00   1.092 0.275886
Visit        -6.045e-03  5.313e-02  -0.114 0.909503
MR.Delay     -1.061e-05  7.774e-05  -0.136 0.891533
Age          -4.208e-03  2.893e-03  -1.454 0.147055
EDUC         -4.831e-03  9.137e-03  -0.529 0.597440
eTIV         -5.154e-05  7.157e-04  -0.072 0.942644
SES          -5.036e-03  2.304e-02  -0.219 0.827147
Alz01         5.850e-01  5.969e-02   9.802 < 2e-16 ***
nWBV         -2.315e+00  6.463e-01  -3.582 0.000404 ***
ASF          -2.283e-02  9.231e-01  -0.025 0.980288
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.08696557)

    Null deviance: 40.601  on 283  degrees of freedom
Residual deviance: 23.829  on 274  degrees of freedom
AIC: 124.18

Number of Fisher Scoring iterations: 2
```

At the 0.05 significance level, we can observe that the variables of Alz01 and nWBV are significant in the model.

Here we will observe the results from the correlation. The correlation coefficient for “Alz01” and “CDR01” is 0.662. The correlation coefficient for “Alz” and “CDR” is 0.624.

Here we will observe the results from the confusion matrices:

Alz01:

The accuracy percentage is 95.71%

	Actual	
Predicted	0	1
FALSE	60	2
TRUE	1	7

Alz:

The accuracy percentage is 95.71%

	Actual	
Predicted	0	1
FALSE	60	2
TRUE	1	7

CDR01:

The accuracy percentage is 88.57%

	Actual	
Predicted	0	1
FALSE	60	5
TRUE	3	2

CDR:

The accuracy percentage is 70.0%

Predicted	Actual		
	0	0.5	1
FALSE	40	20	1
TRUE	0	3	6

Discussion

Recall that the goal of this exploratory data analysis study was to see if we can accurately predict whether an individual will have Alzheimer's, dementia, or both based on known factors about their health. To summarize, we created models for both "Alz01" and "CDR01", and "Alz" and "CDR". Then we split the data into eighty percent as training data, and the remaining twenty percent as test data. After this, we then used the trained data to create new values and predict whether an individual will have Alzheimer's, dementia, or both based on the variables included. Finally, we found and analyzed correlations and confusion matrices to assess how accurate the predictions are.

When looking at "Alz01" and "Alz", models the variables of Age, Clinical Dementia Rating (CDR), normalize whole brain volume (nWBV) were significant in both models. Observing that "CDR" is a statistically significant predictor variable, we can conclude that dementia and Alzheimer's have a connection to one another. When looking at "CDR01" and "CDR", models the variables of whether a person has Alzheimer's "Alz01" and normalize whole brain volume (nWBV) were significant in both models. Observing that "Alz01" is a statistically significant predictor variable, we can further conclude that dementia and Alzheimer's have a connection to one another. The connection between Alzheimer's and dementia is statistically significant when using models in both directions.

The next component of analysis was the correlation coefficients. The correlation coefficient for "Alz01" and "CDR01" is 0.662. The correlation coefficient for "Alz" and "CDR" is 0.624. Both of these are decently strong, and both are consistent with each other in the direction and strength of the correlation. These results can help to further relay the fact that there is a connection between an individual having Alzheimer's and dementia. It shows that having

certain medical predictor variables can help predict whether an individual will have Alzheimer's, dementia, or both.

Lastly, it is vital that we assess the accuracy of the predictions from the trained data. We will do this by observing the accuracy for the confusion matrices. For "Alz01" the accuracy percentage is 95.71%, for "Alz", the accuracy percentage is 95.71%, for "CDR01", the accuracy percentage is 88.57%, and for "CDR", the accuracy percentage is 70.0%. When observing these, we can see that the Alzheimer's variables were the most accurate. Both the 0-3 coded variable ("Alz") and the binomial coded variable ("Alz01") had 95.71% accuracy. The CDR variables had a lower accuracy percentage. The original variable ("CDR") was 18.57% more accurate than the binomial coded variable ("CDR01"), however, these percentages are not fully sufficient. Considering changing the coding of both the CDR variables may help to increase the percentages.

In conclusion, we can observe that there are statistical connections between Alzheimer's and dementia in patients. Additionally, with minor modifications for the "CDR01" and the "CDR" variables, we can accurately predict whether an individual will have Alzheimer's, dementia, or both based on known factors about their health.

References

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